DIPLOMA ELECTRONICS / ECE • ELECTIVE • 60 PERIODS

Optical Communication and Networking

A full handbook for Course Code **5043A** normalized as lesson file **5043**. This is written as a complete learning page: theory, formulas, diagrams, tables, Malayalam support notes, solved problems, revision, model paper and answer key. It is designed to fill the desktop screen instead of sitting as a narrow notebook page.

CREDITS
4

PERIODS
60

CATEGORY
Program Elective

LEVEL
Understanding / Applying

Course Objectives and Outcomes

The syllabus objective is to build conceptual knowledge of optical fibre communication and networking, including signal propagation, components and devices used in modern telecom systems.

Objectives

- ✓ Discuss optical communication as a major part of modern telecom industry.
- ✓ Explain optical fibre communication and networking concepts.
- ✓ Familiarize propagation, optical sources, detectors and network devices.

Course Outcomes

CO	Outcome	Hours	Level
CO1	Outline the concept of light propagation in optical fibre.	14	Understanding
CO2	Explain optical sources and optical detectors.	16	Understanding
CO3	Describe optical transmission and reception.	14	Understanding
CO4	Summarize optical fibre cables and connectors in networking.	14	Understanding
	Series Tests	2	-

Prerequisite Connection

Required background: Applied Physics I and II for fundamentals of light, and Principles of Electronic Communication for modulation, communication blocks and signal concepts. In simple terms: first understand how signals are generated, then learn how the same information is carried by light through a fibre.

> □□□ communication subject □□□□□□□□□□ main idea □□□□ simple □□□: copper wire-□ current □□□□□□□□□ □□□□ fibre-□ light pulse □□□□□□□□. Loss, dispersion, source, detector, connector, network planning □□□□□□□□□□ exam-□ □□□□□□ □□□□□□□.

Official Syllabus Map

This map follows the uploaded 5043A syllabus structure and expands each line into learnable handbook content.

Module	Syllabus Units	Handbook Expansion	Hours
Module 1	Fundamentals of optics, fibre structure, TIR, numerical aperture, acceptance angle, refractive index, fibre types, materials, advantages and applications.	Core optics, calculations, comparison tables, fibre selection logic, industrial examples.	14
Module 2	LED, laser diode, absorption, emission, population inversion, stimulated emission, PIN and avalanche photodiode.	Source/detector working, radiation pattern, modulation, comparison, practical selection.	16
Module 3	Optical system, optical amplifiers SOA/Raman/EDFA, transmitter, receiver, losses, dispersion, transceivers, WDM/DWDM.	Link design, block diagrams, loss budget, dispersion interpretation, multiplexing notes.	14
Module 4	Splicing, joining, connectors, couplers, isolators, circulators, beam splitters, optical modulators, WDM networks, broadcast-and-select, wavelength routed networks, SDH/SONET.	Installation and networking handbook, device role, maintenance checks, network comparison.	14

MODULE 1 • 14 HOURS • CO1

Light Propagation in Optical Fibre

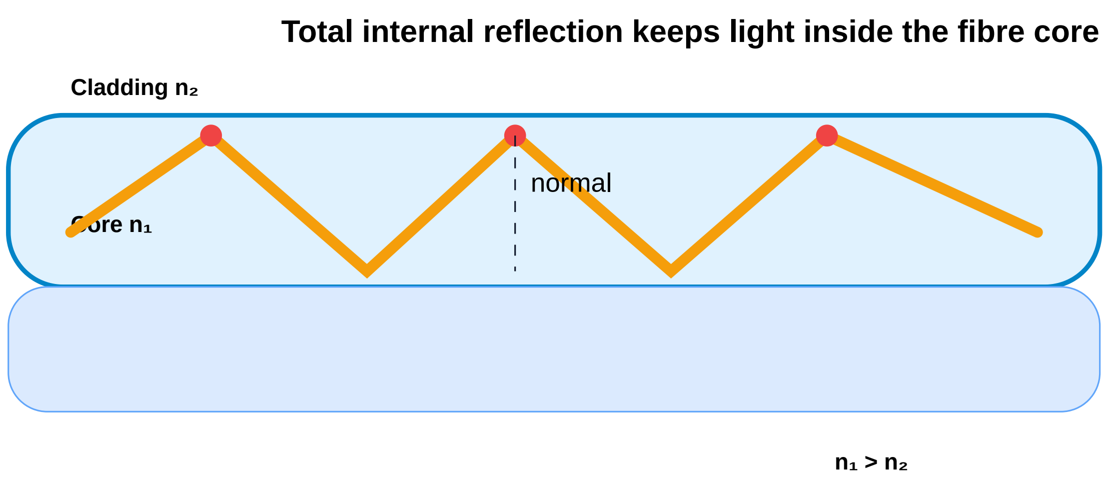
Target: understand why light stays inside the fibre, how acceptance angle and numerical aperture are calculated, and how fibres are classified.

1.1 Fundamentals of Optics

Optical communication uses electromagnetic waves in the optical region. Instead of carrying information as voltage/current through copper, the transmitter converts the information into light intensity variations or optical pulses. The fibre guides this light with very low loss.

- ✓ **Reflection:** light returns to the same medium after striking a surface.
- ✓ **Refraction:** light bends when it travels from one medium to another with different refractive index.
- ✓ **Refractive index:** n = speed of light in vacuum / speed of light in medium.
- ✓ **Critical angle:** angle at which refracted ray travels along the interface.
- ✓ **Total internal reflection:** when incidence angle is greater than critical angle, light reflects completely back into denser medium.

> TIR □□□ fibre communication □□□□ heart. Core refractive index cladding-□□□□□□□□ □□□□□ □□□□□□□□□□□□□□□ light fibre core-□ trap □□□□□□□□□□□□□□□□□.



Principle of light transmission in optical fibre using total internal reflection.

1.2 Fibre Structure

A practical fibre is not just a glass thread. It normally contains:

1. **Core:** central light-guiding region. Refractive index is highest.
2. **Cladding:** surrounds core and provides total internal reflection condition.
3. **Primary coating:** protective polymer layer against moisture and micro-bending.
4. **Strength member and jacket:** mechanical protection for cable installation.

1.3 Numerical Aperture and Acceptance Angle

The fibre accepts only rays entering within a particular cone. The half-angle of this cone is the acceptance angle. Numerical aperture indicates light gathering capacity.

$$NA = \sin \theta_a = \sqrt{n_1^2 - n_2^2} \text{ for air outside the fibre}$$

$$\text{Critical angle: } \sin \theta_c = n_2 / n_1$$

Higher NA means the fibre can accept more light, but it may also allow more modes and create more dispersion in multimode systems.

1.4 Fibre Classification

Basis	Type	Meaning	Use
Transmission mode	Single mode fibre	Very small core, one main light path, low dispersion.	Long distance telecom, backbone links, high bandwidth.
Transmission mode	Multimode fibre	Larger core, many light paths, more modal dispersion.	Short distance LAN, campus links, instruments.
Refractive index profile	Step index	Sudden change between core and cladding.	Simple construction, higher modal dispersion in multimode.
Refractive index profile	Graded index	Core index gradually decreases from centre to edge.	Reduces modal dispersion in multimode fibre.
Material	Glass fibre	Low attenuation and high stability.	Telecom, industrial long links.
Material	Plastic optical fibre	Flexible and easy to handle, but higher loss.	Short links, consumer and automotive uses.

Exam Focus for Module 1

- ✓ Draw fibre structure and label core, cladding and coating.
- ✓ Explain total internal reflection with conditions.
- ✓ Solve numerical aperture and acceptance angle problems.
- ✓ Compare single mode vs multimode and step index vs graded index.
- ✓ Write advantages: high bandwidth, low loss, EMI immunity, small size, security and light weight.

MODULE 2 • 16 HOURS • CO2

Optical Sources and Optical Detectors

Target: select the correct light source and detector for a fibre link and explain their working clearly.

2.1 Optical Sources

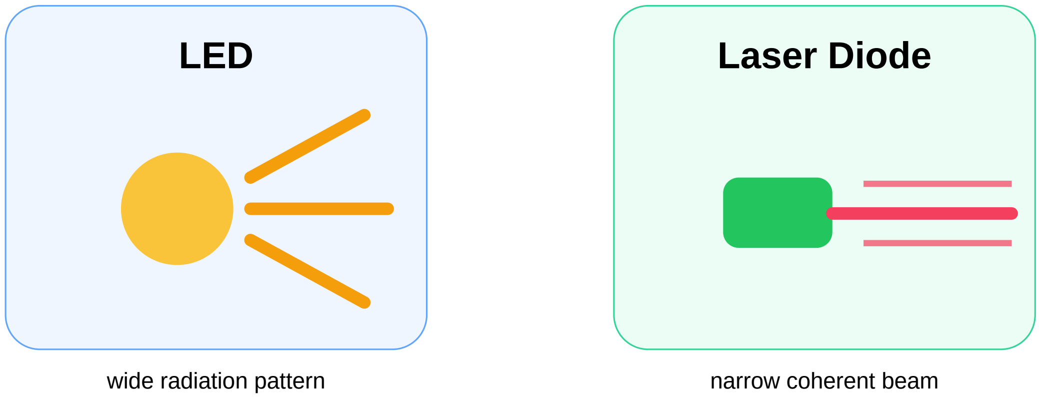
An optical source converts electrical signal energy into optical energy. In fibre communication, source selection is critical because the source must couple light efficiently into the fibre and must support the required modulation speed.

LED

Light Emitting Diode gives spontaneous emission. It is cheaper, reliable and simple, but optical power and spectral purity are lower than laser diode.

Laser Diode

Laser diode gives stimulated emission. Output is coherent, narrow spectral width and suitable for high-speed long-distance links.



LED has wider output; laser diode gives narrow high intensity optical beam.

2.2 LED Types

- ✓ **Surface emitting LED:** light leaves perpendicular to junction plane; simple and commonly used with multimode fibre.
- ✓ **Edge emitting LED:** light leaves from edge; better coupling and higher modulation bandwidth than surface emitting LED.
- ✓ **LED modulation:** optical output follows drive current within safe operating region.

LED short distance link-□□□□□□□ □□□□□□□. Cost □□□□□□□□. □□□□□□ bandwidth□□□ power□□□ laser diode □□□□□ □□□□□□□□□□□□.

2.3 Laser Action

Laser action depends on three important processes:

1. **Absorption:** an electron absorbs energy and moves to a higher level.

2. **Spontaneous emission:** electron returns naturally and emits a photon randomly.

3. **Stimulated emission:** incident photon forces an excited electron to emit another identical photon.

4. **Population inversion:** more electrons exist in excited state than lower state, enabling amplification of light.

2.4 Optical Detectors

An optical detector converts received light back into electrical current. The main requirement is high sensitivity, fast response, low noise and suitable wavelength response.

PIN Photodiode

Uses p-type, intrinsic and n-type layers. It is simple, low noise and commonly used in receivers.

Avalanche Photodiode

Uses avalanche multiplication. It gives internal gain and higher sensitivity, but requires higher reverse bias and has more noise.

2.5 PIN vs APD

Point	PIN	APD
Internal gain	No significant internal gain.	High gain by avalanche multiplication.
Bias voltage	Lower reverse bias.	Higher reverse bias.
Noise	Lower noise.	Higher multiplication noise.
Cost	Lower.	Higher.
Application	General digital fibre receivers.	Weak signal and longer reach links.

Exam Focus for Module 2

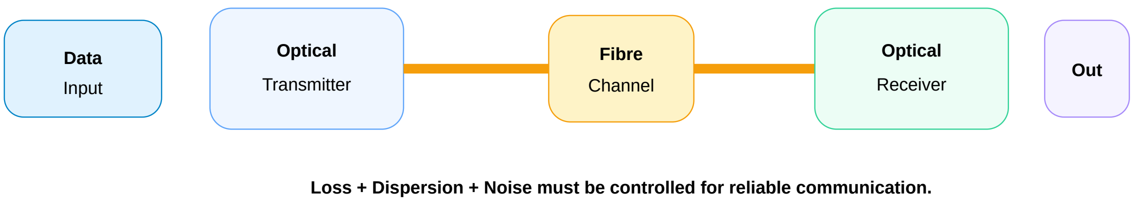
✓ Explain LED structure, surface emitting and edge emitting LEDs.

- ✓ Explain laser diode structure and radiation pattern.
- ✓ Define absorption, emission, stimulated emission and population inversion.
- ✓ Explain photo detection principle.
- ✓ Compare PIN photodiode and avalanche photodiode.

MODULE 3 • 14 HOURS • CO3

Optical Transmission and Reception

Target: understand how a complete optical link works, where losses happen, how amplifiers help, and how WDM/DWDM increases capacity.



Basic optical communication system block diagram.

3.1 Optical Amplifiers

An optical amplifier boosts optical signal power without converting the signal into electrical form. This is important in long fibre routes where regeneration at every stage would be expensive.

Amplifier	Working Idea	Typical Use
SOA	Semiconductor optical amplifier uses active semiconductor medium.	Compact amplification and integrated optical circuits.
Raman	Uses stimulated Raman scattering in fibre itself.	Distributed amplification in long-haul systems.
EDFA	Erbium doped fibre amplifies around 1550 nm window.	Major amplifier in WDM long-distance networks.

3.2 Transmission Losses

- ✓ **Absorption loss:** optical power is absorbed by material impurities or atomic mechanisms.
- ✓ **Scattering loss:** power is scattered due to microscopic variations; Rayleigh scattering is important.
- ✓ **Bending loss:** sharp bend or micro-bend causes light to leak out of the guided path.
- ✓ **Connector/splice loss:** caused by misalignment, air gap, dirty end face or poor polishing.

> Field installation-□ fibre bend radius □□□□□□□□□□□□□□ □□□□ practical mistake □□□. Optical cable copper cable □□□□ □□□□□□□□ □□□□□□□□□□.

3.3 Dispersion

Dispersion spreads optical pulses in time. If spreading is high, neighbouring pulses overlap and cause inter-symbol interference.

- ✓ **Intermodal dispersion:** different modes travel different paths; occurs in multimode fibre.
- ✓ **Intramodal dispersion:** different wavelengths travel at different speeds; includes material and waveguide dispersion.
- ✓ **Effect:** limits maximum bit rate and transmission distance.

3.4 Optical Transmitter and Receiver

Transmitter: encoder/driver controls LED or laser diode and couples the optical output into fibre. **Receiver:** detector converts light into current, followed by pre-amplifier, decision circuit and output interface.

1. Electrical signal is shaped and encoded.
2. Optical source is modulated.
3. Light travels through fibre and experiences attenuation/dispersion.
4. Photodetector converts received optical power into current.
5. Receiver electronics recover the transmitted data.

3.5 WDM and DWDM

Wavelength Division Multiplexing sends several optical channels through the same fibre using different wavelengths. Dense WDM uses closer channel spacing and supports a larger number of channels.

Total fibre capacity ≈ number of wavelengths × bit rate per wavelength

Feature	WDM	DWDM
Channel spacing	Wider	Narrower
No. of channels	Lower	Higher
System cost	Moderate	High
Application	Metro/simple capacity expansion	Long-haul and backbone networks

MODULE 4 • 14 HOURS • CO4

Optical Cables, Connectors and Networking

Target: understand how optical components are joined, split, protected, modulated and arranged into networks.

4.1 Fibre Splicing and Joining

Splicing joins two fibre ends permanently or semi-permanently. Good splicing requires clean end faces, correct alignment and minimum insertion loss.

Method	Idea	Result
Fusion splicing	Fibre ends are aligned and fused using electric arc.	Low loss and high reliability.
Mechanical splicing	Fibre ends are aligned in a mechanical holder with index matching gel.	Quick repair, usually higher loss than fusion.
Connectorized joint	Detachable connection using connector pair and adapter.	Useful for patch panels and equipment ports.

4.2 Connector Handling Rules

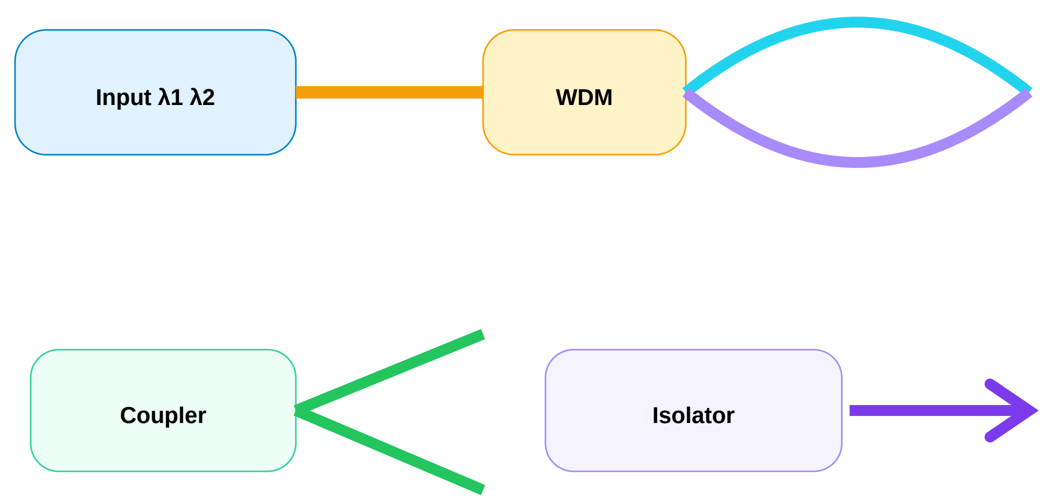
- ✓ Never touch polished fibre end face with fingers.
- ✓ Keep dust caps fitted until connection time.
- ✓ Clean connector using approved fibre cleaning tools.
- ✓ Check bend radius and strain relief.
- ✓ Record link loss after installation.

Connector dirty □□□□□□□□□ optical power □□□□□□□. Electrical terminal loose

□□□□□□□□□□□□□ fibre connector contamination □□□ common fault.

4.3 Passive and Active Optical Components

- ✓ **Coupler:** combines or splits optical power.
- ✓ **Directional coupler:** transfers controlled power between waveguides/fibres.
- ✓ **Beam splitter:** divides optical beam into two or more beams.
- ✓ **Optical isolator:** allows light in one direction and blocks reverse reflection.
- ✓ **Circulator:** routes light from port 1 to 2, 2 to 3 and so on.
- ✓ **Optical modulator:** varies optical signal according to information signal.



Optical networking uses multiplexers, couplers, isolators, circulators and splitters.

4.4 Optical Networks

Optical networking uses fibre links and wavelength channels to connect many nodes. The syllabus highlights WDM networks, broadcast-and-select networks, wavelength-routed networks and SDH/SONET concepts.

Network Concept	Meaning	Important Point
Broadcast-and-select network	Signal is broadcast to many nodes and receiver selects required wavelength/channel.	Simple sharing concept, but power is split and scalability is limited.
Wavelength-routed network	Optical path is created using wavelength routing through network nodes.	Better for large networks and efficient wavelength reuse.
SDH/SONET	Synchronous optical transport hierarchy used in telecom backbone.	Provides standard frame structure, multiplexing and management.

Field-Oriented Maintenance View

In real installations, optical problems are usually caused by connector contamination, bad bend radius, poor splicing, wrong patching, optical power mismatch, transmitter failure or receiver sensitivity limit. A technician should test continuity, connector cleanliness, transmit power, receive power and link loss before blaming the electronics.

Formula Bank and Quick Facts

Use this section for fast revision and numerical problems.

Optics Formulas

Refractive index: $n = c / v$

Critical angle: $\sin \theta c = n_2 / n_1$, where $n_1 > n_2$

Numerical aperture: $NA = \sqrt{(n_1^2 - n_2^2)}$

Acceptance angle in air: $\theta_a = \sin^{-1}(NA)$

Fractional index difference: $\Delta = (n_1 - n_2) / n_1$

Link Formulas

Loss in dB = $10 \log_{10}(Pin / Pout)$

Pout(dBm) = Pin(dBm) - total loss(dB)

Total link loss = fibre loss + splice loss + connector loss + margin

Capacity with WDM = channel count × bit rate per channel dB calculations exam-□ □□□□□□□□□ sign □□□□□□□□□□□□□□. Loss subtract □□□□□□□□; gain add □□□□□□□□.

Solved Examples

Representative examples based on syllabus formulas.

Example 1: Numerical Aperture

Core refractive index $n_1 = 1.48$ and cladding refractive index $n_2 = 1.46$. Find NA.

$$NA = \sqrt{(1.48^2 - 1.46^2)} = \sqrt{(2.1904 - 2.1316)} = \sqrt{0.0588} = 0.2425$$

Answer: NA ≈ **0.243**.

Example 2: Acceptance Angle

If NA = 0.25, find acceptance angle in air.

$$\theta_a = \sin^{-1}(0.25) = 14.48°$$

Answer: acceptance angle ≈ **14.5°**.

Example 3: Fibre Link Loss

A 20 km link has fibre attenuation 0.35 dB/km, 4 splices of 0.1 dB each, 2 connectors of 0.5 dB each and margin 3 dB. Find total loss.

$$\text{Fibre loss} = 20 \times 0.35 = 7 \text{ dB}$$

$$\text{Total} = 7 + 0.4 + 1 + 3 = 11.4 \text{ dB}$$

Answer: total link loss = **11.4 dB**.

Example 4: WDM Capacity

A WDM system carries 16 wavelengths and each wavelength carries 2.5 Gb/s. Find total capacity.

$$\text{Capacity} = 16 \times 2.5 = 40 \text{ Gb/s}$$

Answer: **40 Gb/s**.

Practice Question Bank

Questions are arranged for short answer, explanation and comparison practice.

Module 1 Questions

Q1 Define total internal reflection and state the conditions.

Q2 Draw and explain the structure of optical fibre.

Q3 Define numerical aperture and acceptance angle.

Q4 Compare single mode and multimode fibre.

Q5 Explain step index and graded index fibres.

Q6 List advantages and applications of optical fibre.

Module 2 Questions

Q7 Explain surface emitting LED and edge emitting LED.

Q8 Explain laser action with stimulated emission.

Q9 Draw the structure of laser diode and explain radiation pattern.

Q10 Explain principle of photo detection.

Q11 Explain PIN photodiode.

Q12 Compare PIN and avalanche photodiodes.

Module 3 Questions

Q13 Draw optical communication system block diagram.

Q14 Explain SOA, Raman amplifier and EDFA.

Q15 List and explain optical fibre losses.

Q16 Explain intermodal and intramodal dispersion.

Q17 Draw optical receiver block diagram.

Q18 Explain WDM and DWDM.

Module 4 Questions

Q19 Explain fibre splicing and joining methods.

Q20 Explain directional coupler and its applications.

Q21 Explain optical isolator and circulator.

Q22 Explain beam splitter and optical modulator.

Q23 Compare broadcast-and-select and wavelength-routed networks.

Q24 Write short note on SDH/SONET.

Model Question Paper with Answers

Practice model based on the syllabus weightage. Use it for exam rehearsal; it is not a copied official question paper.

Course: 5043 Optical Communication and Networking
Time: 3 Hours   Maximum Marks: 100

PART A - Short Answer Questions
1. Define refractive index.
2. State the condition for total internal reflection.
3. Define numerical aperture.
4. Name two types of optical sources.
5. What is population inversion?
6. Define photo detection.
7. What is fibre bend loss?
8. What is WDM?
9. What is fusion splicing?
10. What is an optical isolator?

PART B - Answer any eight
11. Explain optical fibre structure with labelled diagram.
12. Compare single mode and multimode fibre.
13. Derive the expression for numerical aperture.
14. Compare LED and laser diode.
15. Explain PIN photodiode and APD.
16. Explain optical communication system block diagram.
17. Explain optical amplifiers: SOA, Raman and EDFA.
18. Explain absorption, scattering and bending losses.
19. Explain WDM and DWDM.
20. Explain optical couplers and beam splitters.

PART C - Long Answer Questions

21. Explain total internal reflection and propagation of light in optical fibre. Add numerical aperture and acceptance angle.
22. Explain optical sources and detectors with neat comparison tables.
23. Explain optical transmitter, fibre channel and receiver with losses and dispersion.
24. Explain optical networking using WDM, broadcast-and-select and wavelength-routed networks.

Answer Key / Key Points

Part A Key

1. $n = c/v$.
2. Light must travel from denser to rarer medium and incidence angle must be greater than critical angle.
3. NA is light gathering ability of fibre; $NA = \sqrt{(n_1^2 - n_2^2)}$.
4. LED and laser diode.
5. More electrons in excited state than lower energy state.
6. Conversion of optical signal into electrical signal.
7. Loss due to sharp macro/micro bending of fibre.
8. Transmission of many wavelengths through same fibre.
9. Permanent joining of fibre ends by electric arc.
10. Device that passes light in one direction and blocks reverse reflections.

Long Answer Checklist

✓ For diagrams, label blocks clearly and write function of each block.

✓ For comparisons, use table format.

✓ For numerical problems, write formula, substitution and unit.

✓ For networking questions, mention WDM, DWDM, routing and channel selection.

✓ For detector questions, include sensitivity, noise, bias and gain.

Text / Reference and Online Resources

Books

- Fiber Optic Communication - Systems and Components / Vivekanand Mishra and Sunita P Ugale.
- Introduction to Fiber Optics / Ajay Ghatak and K Thyagarajan.
- Optical Fiber Communication / Gerd Keiser.
- Optical Fiber Communication / John M Senior.
- Optical Networks: A Practical Perspective / Rajiv Ramaswamy.

Online Topics to Study

- Optical fibre telecommunication basics.
- Optical communication lecture notes.
- WDM and DWDM tutorials.
- Connector cleaning and fibre link loss budget tutorials.

Final Revision Advice

Do not study this subject only as theory paragraphs. Draw every block diagram repeatedly: fibre structure, TIR, optical transmitter, optical receiver, WDM system, coupler and networking. Then practise NA, acceptance angle, link loss and WDM capacity. That is the shortest route to scoring well.